TOBB ETÜ

University of Economics & Technology

Proposal

Final Stack

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1. Project Title : **LucidPawn**

2. Problem Definition & Objectives

- **Problem Definition:**
  In chess, many players have a hard time keeping track of their moves or recording them for offline analysis. Current tools generally require digital input (PGN/algorithmic notation) or only work through screen capture which do not meet the demand for real-time position extraction, move detection and level/strength based personalized recommendations.

- **Objectives:**
  - Detecting chess piece and square detection through pictures/videos of physical chess board
  - Let users stream their game sessions and store them.
  - Users can share their game sessions and view other publicly shared games.
  - Give viewers feedback about moves, decisions with replay of session options.

3. Scope

1. Project Objective

The project aims to develop an AI-based chess analysis system that automatically captures real-world chessboard positions through computer vision, analyzes them with powerful backend engines such as Stockfish 15 or equivalent UCI-compatible models, and stores complete game histories. The system will provide move recommendations, real-time assistance, and visual or interactive tools for players and instructors.

2. Functional Scope

- **Board Capture:** Detect and extract the chessboard from camera input, correct perspective, segment squares, identify chess pieces, and generate valid FEN representations after each move.

- **Position Normalization and Game Reconstruction:** Convert recognized board states into FEN, generate PGN move lists, and ensure move legality and consistency across frames.

- **Backend Engine Integration:** Use Stockfish 15 or other UCI-compatible engines for evaluation and move suggestions. Allow adjustable parameters such as depth, search time, or number of candidate moves.

- **Game History Management:** Store complete games with FEN and PGN data, timestamps, captured images, and metadata. Support game retrieval, filtering, and export for long-term study.
- **Game Share:** Let user share their live game or saved games.

3. Assumptions and Dependencies

- Adequate lighting and a standard chessboard are assumed for reliable recognition.

- The backend server must support execution of Stockfish or equivalent UCI engines.

- A local or cloud database will be used to store user data, games, and analysis results.

- Data collection and storage will comply with privacy and consent requirements.

4. Deliverables

- **Computer Vision Module** with trained detection and classification model.

- **Backend Engine Service** (REST/GraphQL) for position evaluation, move suggestion, and game storage.

- **Database Schema** for users, games, positions, and analytics.

- **User Interface** with interactive 2D/3D board visualization, evaluation metrics, and real-time feedback.

- **Performance and Evaluation Report** including accuracy, latency, and system scalability metrics.

5. Success and Acceptance Criteria

- Piece and board detection accuracy ≥ X%; move reconstruction accuracy across N consecutive moves ≥ Y%.

- Average evaluation latency ≤ Z seconds per position.

- Full-game import/export in PGN format without data loss.

- Stable operation during real-time training or analysis sessions.
- Target thresholds (X, Y, Z, N) will be defined after the initial prototype phase and progressively increased as the model and system are optimized.

4. User Profile

1. Amateur Chess Players
Individuals who play chess casually and want to improve their strategic thinking by reviewing their own moves.

2. Street Chess Players
People who play chess informally in public spaces. The system can help them analyze their games afterward by capturing moves through images, offering feedback in a simple, accessible way without requiring digital boards.

3. Chess Club and Tournament Players
Competitive players seeking to refine their tactics. The project can support post-game analysis, highlight mistakes, and suggest alternative strategies based on professional-level data.

4. Chess Instructors and Training Centers
Coaches or chess academies that wish to use AI-generated insights to guide students' learning, evaluate progress, and visualize game patterns for educational purposes.

5. Students and AI Enthusiasts
Learners or developers interested in combining artificial intelligence with chess analysis to study both gameplay patterns and practical AI applications.

5. Anticipated Challenges & Constraints

- **Technical hurdles**

1. Image Recognition Accuracy
Accurately detecting chess pieces and their positions from captured images can be technically challenging. Variations in lighting, camera angle, or board design may reduce recognition accuracy and lead to incorrect move analysis.

2. Real-Time Processing
Processing images and generating move suggestions in real time requires optimized algorithms and efficient model design. Delays or lag in prediction could negatively affect user experience and system usability.

3. Model Training Complexity
Developing a reliable recommendation model demands substantial training data and fine-tuning. Ensuring that the model correctly interprets board states and predicts logical next moves involves complex learning processes and parameter optimization.

4. System Integration and Maintenance
Integrating the AI module with the user interface, camera input, and database systems may introduce compatibility and synchronization issues. Continuous updates and testing will be required to maintain stable performance.

- **Time/resources limits**

1. Computational Constraints
Training and evaluating the AI model require significant computational resources, especially for image recognition and deep learning processes. Limited access to powerful hardware, such as GPUs, may slow down experimentation and reduce model performance or efficiency.

2. Data Availability
There are already huge dataset of different kind of matches, game sessions available free.

- **Ethical or regulatory considerations**

1. Data Privacy and Security
User data, including images of chess moves, must be collected and stored responsibly. Players should give explicit consent before any data recording. All stored information must be encrypted and accessible only to authorized personnel. The system should comply with privacy regulations, and users should be informed about how their data is processed, used, and deleted.

2. Fair Play and Cheating Prevention
Since the system provides move recommendations, it must not encourage or enable unfair

advantages in competitive or online matches. The tool should clearly state that its purpose is educational or analytical, not for real-time game assistance during official play.

6. Data Sources

There are collections of recorded games, often including professional, amateur, and historic games. They help engines learn opening theory, middlegame patterns, and endgame ideas. These are:

- FICS Games Database
- Lichess Database
- Chess.com Database